Ordered-n-tuple:: Specific order $n=3$ $(5,7,9) \neq (7,5,9)$

Two ways to write a vector::

1) comma form (row) $(x_1, x_2, x_3 \cdots x_n)$

2) column form $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

Transformation:: Takes in a vector and gives you another vector.

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

↓

T is a n-dim vector that produces a m-dim vector.

1) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ input (x,y) output $(x+y, 2x, 3y)$

if n=m then we call T a operator

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$T(x,y,z) = (2x, 3y, 4z)$

Q1 $\mathbb{R}^4 \rightarrow \mathbb{R}^3$

x_1, x_2, x_3, x_4

↓

4d

$w_1 = 2x_1 - 3x_2 + x_3 - 5x_4$

$w_2 = 4x_1 + x_2 - 2x_3 + x_4$

$w_3 = 5x_1 - x_2 + 4x_3$

$$\begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix}$$

Input vector given

$x = \begin{pmatrix} -1 \\ 3 \\ 0 \\ 2 \end{pmatrix}$

$$\begin{bmatrix} 2 & -3 & 1 & -5 \\ 4 & 1 & -2 & 1 \\ 5 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \\ 0 \\ 2 \end{bmatrix}$$

$2(1) + (-3)(-3) + 1\times0 + 2\times-5$

$4(1) - 3 - 2\times0 + 1\times2$

$5\times1 - 3\times(-1) + 4\times0 + 0\times2$

$$= \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$$

Zero transformation:: Zero vector $\times$ any vector $= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Identity operator:: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $T_I(x) = I \cdot x$

If we multiply any vector by the identity matrix you get the same vector back.

Rules::

→ $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $k = 3$

→ Transform the multiply

$T_A(ku) = k\,T_A(u)$

$T_A(3u) = T_A\begin{bmatrix} 3 \\ 6 \end{bmatrix}$

$3\,T_A\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

→ transform then add

$T_A(u+v) = T_A(u) + T_A(v)$

Add first then transform

$u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$$\begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$\rightarrow T_A(u-v) = T_A(u) - T_A(v)$

linear trasformation is the same as matrix trasformation

$T(k \cdot u) = k \cdot T(u)$

$T(u+v) = T(u) + T(v)$

↓ JUST multiply a vector by a matrix

$T(x) = A \cdot x$

$T(x,y) = A\begin{pmatrix} x \\ y \end{pmatrix}$

$T(ku) = k\,T(u)$

$k=2 \quad u=(1,2)$ $\qquad T(x,y) = (2x+3y,\ x-y)$

$$T(2,4) = (2(2)+3(4),\ 2-4)$$
$$= (4+12,\ -2)$$
$$= (16,-2)$$

$kT(u) \rightarrow 2\left[2+3(2),\ 1-2\right] \rightarrow 2(8,-1) \rightarrow (16,-2)$

$T(x,y) = A\begin{bmatrix} x \\ y \end{bmatrix}$ $\quad T(x,y) = (2x+3y,\ x-y)$

$$A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$$

$$T(x,y) = \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

(Basis of a vector space)

↓ used to build vector

Standard Basis

$e_1 = (1, 0, 0, \cdots 0)$
$e_2 = (0, 1, 0 \cdots 0)$
$e_3 = (0, 0, 1 \cdots 0)$

e.g in $\mathbb{R}^3$

$e_1 = (1,0,0)$
$e_2 = (0,1,0)$
$e_3 = (0,0,1)$

Finding the Standard Matrix Transformation

Standard matrix

$$A = [T(e_1) \mid T(e_2) \mid T(e_n)]$$

$T(x,y) = (2x+3y,\ x-y)$

Standard basis in $\mathbb{R}^2$ $\quad e_1 = (1,0)$ $\quad e_2 = (0,1)$

$T(e_1) = T(1,0) = (2(1)+0,\ 1-0)$

$= (2,1)$ $\qquad T(e_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$T(e_2) = T(0,1) = (2(0)+3,\ 0-1)$

$= (3,-1)$ $\qquad T(e_2) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

$\begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$ Standard matrix of transformation.

→ $T: \mathbb{R}^2 \to \mathbb{R}^3$ (T takes a 2-number vector and outputs 3-number vector)

→ Ax (A multiplies by x)

this is how transformation is done

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad x = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

$$Ax = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix}$$

→ Identity matrix

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad Iu = u$$

→ $T_I(u) = u$ ($x \to$ is a vector)

→ $T(x) = Ax$ (linear transformation)

e.g 7, e.g 8

For any angle θ, the rotation matrix is always

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \to 2D$$

Find the image of $x = (1,1)$ under a rotation of $\pi/6$ radians (30°) about the origin.

$$R_\theta = \begin{bmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{bmatrix}$$

$$R_{\pi/6} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$ → now multiply this by any vector to rotate that vector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$$\begin{pmatrix} \frac{\sqrt{3}}{2} - \frac{1}{2} \\ \frac{1}{2} + \frac{\sqrt{3}}{2} \end{pmatrix}$$ → final Ans